Predicting Singapore HDB Rents with Comparable Transactions and Temporally Diverse Ensembles

Han Linxuan A0350911M
Sun Shenghe A0354849L
Zhang Yuhong A0356795H
Zhang Zhexuan A0355290A

***Abstract*—We investigate monthly rental prediction for Singapore Housing and Development Board flats using 150,000 labelled transactions. The main challenges are market change, heterogeneous flats and the absence of reliable individual-flat identifiers. Consistent preprocessing and spatial joins support comparable-transaction statistics; market adjustment and training-history augmentation then provide complementary inputs to an eight-component CatBoost ensemble. On January–March 2025 it achieves RMSE 440.76 and MAE 316.55 SGD, versus RMSE 445.52 for a four-component market-adjusted reference. Gains also appear in two longer development windows, but an older annual window worsens by 9.81 RMSE. Transactions at or below 2,500 SGD contribute 52.23% of recent squared error despite representing only 16.34% of observations. Further model and information tests do not establish a stable replacement. The study therefore documents both useful comparable-market information and the limits of transferring recent gains across market periods.**

***Index Terms*—HDB rental prediction, tabular regression, comparable transactions, temporal validation, ensemble learning**

I. Introduction

Monthly rent predictions can help renters compare options and owners assess asking prices. Singapore's public housing market requires considering physical attributes, surroundings and changing market conditions together. We estimate recorded monthly rent from observable transaction characteristics and investigate where those estimates remain unreliable.

The course provides a regression task built from HDB rental approval information and auxiliary datasets. We treat it as dated cross-sectional transaction prediction. The underlying process may contain repeated observations of individual flats, but the released fields do not establish those identities. Several flats in a block can share their type, model and area. A block-level rental history is therefore evidence about comparable transactions, not a verified trajectory for one flat. This distinction matters both when handling repeated records and when interpreting historical features.

Three research questions organise the study. *RQ1:* What information is recovered through consistent preprocessing and spatial joins? *RQ2:* Do market-adjusted comparable rents and different training-history views improve local prediction? *RQ3:* Do these improvements transfer across periods, and which groups account for the remaining errors? Local chronological evaluation and interpretation are our primary evidence; competition scores provide a supplementary external check.

CS5228 Group 44, National University of Singapore. Results include follow-up analysis through 5 November 2026.

Our empirical contribution combines auditable preprocessing, a comparable-market predictor with training-history diversity, and temporal and subgroup evaluation that retains adverse findings. Code and cumulative experimental records are maintained in the team repository.[1]

II. Data Understanding and Preparation

A. Released files and task boundaries

The training file contains 150,000 rows from January 2021 through March 2025; the unlabelled competition file contains 50,000 rows from March 2025 through July 2026. Ten inputs describe approval month, town, block, street, flat type, flat model, floor area, furnishing status, lease commencement year and fee; training additionally supplies monthly rent. These audited file counts and dates take precedence over the course webpage's broader dataset overview.

Month bins overlap in March 2025: 3,032 training rows and 380 competition rows occur in that month. Thus the competition period is predominantly later, but month precision does not establish strict chronological separation for every transaction. Competition rent labels are not supplied; no modeling experiment or submission uses public-source test-period rental targets. Approval month is not necessarily the month in which rent was negotiated, limiting interpretations of short-term market movement.

Training rent has mean 2,719.04, median 2,700, standard deviation 749.73, and range 300–7,600 SGD. No core input is missing, although complete fields do not establish correctness or complete economic information. Auxiliary tables describe blocks, MRT stations, malls, primary schools, Certificate of Entitlement (COE) prices and stock prices. A normalised block lookup covers every released rental row without missing coordinates or principal block attributes.

B. Cleaning guided by exploratory analysis

We lowercase categories, trim whitespace, replace punctuation with spaces and collapse repeated spaces. For example, `3-room` and `3 room` become the same category. Raw flat-type labels reduce from eleven to six and street labels from 1,146 to 576 in training. Identical deterministic rules apply to training and competition inputs. This prevents formatting variants from fragmenting comparable groups while preserving immutable raw CSVs.

[1] https://github.com/ElephantLikeCode/cs5228-hdb-rental-prediction

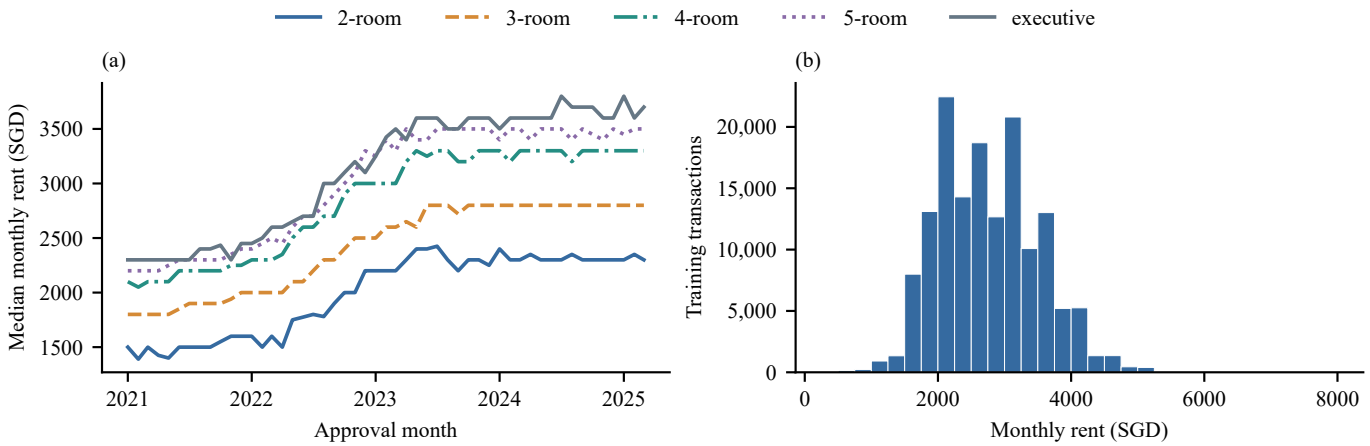


Fig. 1. Training-data profile. Left: monthly median rent by canonical flat type; one-room flats are omitted because there are only 42 training observations. Right: the distribution of all 150,000 training rents. Temporal curves describe changing transaction samples, not repeat-flat or causal price indices.

Furnishing status and fee are constant and are dropped. Calendar year, month, quarter, a month index and nonnegative lease age are derived. Floor area is in square metres; comparable-area keys round it to 0.1 square metres. Lease age uses the recorded lease commencement year rather than renovation condition or tenancy duration. Address combines normalised block and street. Block joins also use town and enforce many-to-one matching, preventing accidental row multiplication.

CatBoost receives canonical strings explicitly identified as categorical columns. Linear and Ridge baselines use training-fitted one-hot encoding with unknown values ignored, median imputation and numeric scaling. The scikit-learn tree baselines use training-fitted ordinal encoding, an unknown-category sentinel and numeric median imputation. XGBoost uses native categorical splits with vocabularies fitted on training only. These representations and model capacities differ, so the baseline comparison does not isolate algorithm choice alone.

Fig. 1 motivates two subsequent choices. Monthly medians change markedly, making time-invariant group averages inadequate as the only predictor. Flat-type differences also persist, motivating market normalisation within flat type rather than pooling all room counts into one index. Neither descriptive pattern identifies a causal effect, because transaction composition changes over time.

C. Repeated inputs and extreme rents

An initial audit finds 425 exactly repeated labelled rows. Equal recorded fields and rent are not sufficient evidence of the same flat or contract, so these observations remain. Normalised feature vectors can also repeat with substantially different rents. Removing such records would change the frequency of transaction types without resolving absent individual-flat identity.

Attribute tracing strengthens this caution: all 13,758 recurrent building/type groups have constant supplied area, model and lease year, whereas historical resales allow multiple areas within a group. The supplied attributes therefore need not identify an individual rental unit. Compatibility with resale records does not recover the course's exact join rule or justify assigning a resale floor to a particular tenancy.

Low rents for larger flats motivate a sensitivity study. A fixed rule flags three-room-or-larger or executive observations below half the median for the same approval month, town and canonical flat type. It flags 954 training rows, or 0.636%. Such a target-defined flag is a diagnostic or training-policy intervention; it cannot be computed from an unlabelled test transaction. We evaluate alternative training policies on the same complete local validation population and preserve every competition row in its original order. We neither label all low rents as errors nor multiply them by a presumed room count.

III. Methods

A. Spatial information and baseline learners

Block features include coordinates, maximum floor, completion year, subzone, planning area and region. Great-circle distances connect each block to the nearest MRT station, mall and primary school; counts within 0.5, 1 and 2 km describe nearby amenities. These are geographic proxies rather than walking or transit travel times. Predictive association with proximity must not be described as a causal premium.

Global and town/flat-type medians establish simple references. Linear Regression, Ridge, Decision Tree, Random Forest [1] and histogram gradient boosting span increasingly flexible approaches. CatBoost [2] handles numeric and high-cardinality categorical inputs and is selected empirically. Later, matched-input XGBoost [3] tests learner complementarity. These comparisons do not establish universal algorithm superiority.

B. Past-only comparable statistics

For query month t, historical features use only permitted training transactions strictly earlier than t. No query-month

rent, including a training row's own rent, enters its statistics. For group g, an initial smoothed mean is

$$\widehat{\mu}_{g,t} = \frac{\sum_{i \in H_{g,t}} y_i + \lambda \mu_t}{|H_{g,t}| + \lambda}, \quad \lambda = 20, \tag{1}$$

where $H_{g,t}$ is the earlier group history and μ_t the earlier global mean. Log support counts accompany these means. Groups include town/type, address/type, address/type/rounded area, street/type and town/type/model, as well as broader groups. These keys identify comparables, not unique flats. Empty early history remains missing with explicit support information; adaptive levels use a fixed 2,000 SGD cold-start prior.

Unadjusted historical rents confound local premium with market movement. Our adaptive representation divides each reference rent by its own flat-type/month mean. It estimates decayed premiums and shrinks specific groups toward parents: town/type, town/street/type, address/type, and address/type/area. Month indices use $m = 12\,\mathrm{year} + \mathrm{month}$, so differences measure elapsed months. Weights $w_i = 2^{(m_i - a)/12}$ are anchored at the latest permitted reference month a, giving

$$\widehat{r}_{g,t} = \frac{\sum_{i \in H_{g,t}} w_i r_i + \alpha_g \widehat{r}_{p(g),t}}{\sum_{i \in H_{g,t}} w_i + \alpha_g}. \tag{2}$$

Here r_i is the reference rent ratio, the top-level parent premium is one, and the four shrinkage strengths are 30, 12, 8 and 5. Anchoring weights to the latest observation avoids making effective support vanish merely because the query is distant. Each group supplies its market-adjusted level, premium, effective support and observation age.

For flat type f, let $M_{f,a}$ be the count-weighted mean of its latest three observed monthly means. Its log-linear slope β_f uses up to twelve months, is zero with fewer than six, and is clipped to $[-0.01, 0.01]$ per month. For horizon $h = t - a$ from the final reference month,

$$\widehat{M}_{f,t} = M_{f,a} \exp\left[\beta_f \, 6(1 - e^{-h/6})\right]. \tag{3}$$

Multiplying this forecast by (2) gives a comparable-market level. Damping limits extrapolation but does not eliminate regime risk. These coefficients are recorded design choices rather than parameters fitted to outer validation labels.

C. Local trends and training-history diversity

Four town/flat-type features capture movement relative to the flat-type market: normalised premium in the latest six reference months, premium in the preceding six, their log ratio clipped to $[-0.15, 0.15]$, and log support based on the smaller period count. Both period ratios shrink toward one with strength 20. Thus the method measures local relative momentum rather than fitting an unconstrained future trend independently for each block.

A second training view simulates stale information. Eligible observations are copied with the latest six months of reference labels hidden, while preserving their actual query dates. Eligibility requires at least twelve months since the first reference month. The original and copy each receive half the observation's original weight, preserving total weight. This creates alternative information conditions, not new independent rental evidence. At inference, all components use the full permitted training history for each query.

Each view has a direct rent model f and a model g for residual $y - b_v(x)$, where $b_v(x)$ is the adaptive address/type/area level. Averaging both target modes over seeds 42 and 2026 yields

$$\widehat{y}(x) = \frac{1}{8} \sum_{v} \sum_{s} \left[f_{v,s}(x) + b_v(x) + g_{v,s}(x) \right], \quad v \in \{\mathrm{normal}, \mathrm{stale}\}, \quad s \in \{42, 2026\}. \tag{4}$$

The normal view includes local trends; the augmented view uses hierarchical features without those four trend columns. Component input dimensions are 78 and 74 respectively. Equal weights preserve diversity without fitting another layer to the latest evaluation errors. Fig. 2 makes the component count explicit; "head" below denotes a view/target pair averaged over its two seeds.

D. Alternative hypotheses

Stacking [4] uses predictions generated by earlier training origins to fit a second layer. We compare four seed-averaged heads with all eight individual outputs, using constrained weighting or Ridge correction. Meta-training always precedes meta-evaluation. The retained probe fits nonnegative sum-to-one four-head weights on October–December 2024 and shrinks them halfway toward equal weights; January–March 2025 labels are not used to fit those weights.

The nearest-comparable predictor finds up to 64 same-type transactions using scaled coordinates, area and lease age. It weights market-normalised rents by inverse squared distance and eighteen-month recency decay, then restores a flat-type market level. The reported variant holds that market level constant after the origin; damped and bounded-linear forecasts are additional checks. Flat-type specialists fit separate CatBoost models for one/two-room, three-room, four-room and five-room/executive groups, falling back to the selected ensemble below 500 training rows. Rent-band experts fit three regressors on training rents $\leq 2{,}500$, $(2{,}500, 4{,}000]$ and $> 4{,}000$ SGD; a CatBoost classifier supplies prediction-time mixture probabilities. Each of these candidates enters a fixed 25% blend with the selected ensemble.

Other tests use five-fold cross-fitted statistics, out-of-fold (OOF) residual corrections and matched-input learners. OOF predictions come from models not fitted on the predicted row's label; corrections learn earlier-origin OOF residuals. An isolated area/age composition-trend ablation preserves original history levels and the residual offset. These definitions distinguish it from an earlier joint change.

Economic experiments use lagged supplied COE and stock summaries, then official CPI and URA private non-landed rental indices. CPI measures consumer prices, while URA measures private housing rather than HDB rents. Release-lagged levels and changes are supplied with missingness and staleness indicators. A separate MRT calendar reconstructs

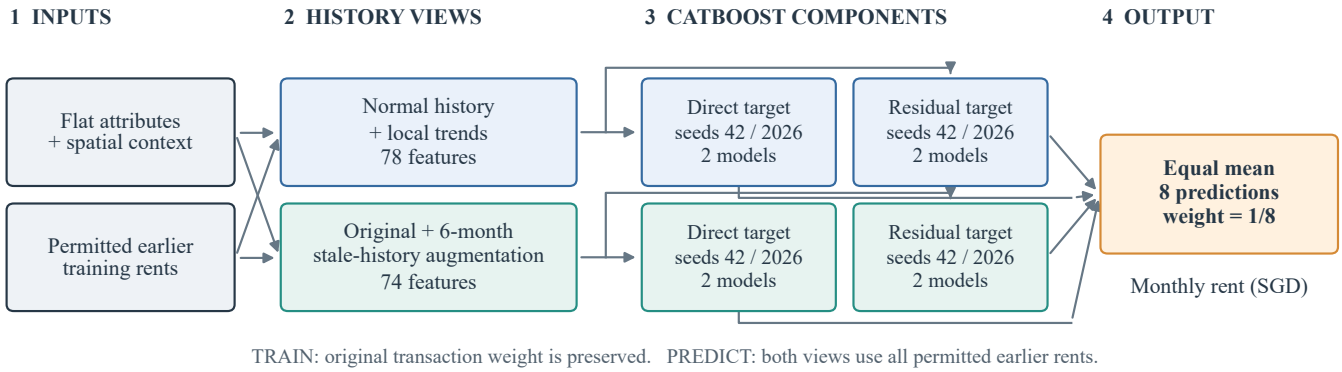


Fig. 2. Selected model, read left to right. Two history views each feed direct and residual CatBoost targets, each fitted with two seeds: $2 \times 2 \times 2 = 8$ components. Residual predictions restore their comparable-rent offset before averaging. Augmentation preserves each original transaction's total weight; inference uses all permitted earlier training rents in both views, not rents from the scoring interval.

operational stations by rental month and deduplicates interchange names. Opening exposure describes distance-decayed new-line access, recent openings and months since opening. These features test predictive association, not a causal opening effect.

A follow-up adds three access proxies to the original local-trend inputs: distance to the nearest operating interchange and distinct supplied line-code prefixes within 1 and 2 km. Service follows the audited calendar; validation admits only openings announced before its origin. Two seed-42, 1,600-tree direct/residual fits per window test replacement of their corresponding ensemble components and a fixed 25% blend. These measures retain the original static inputs and do not reconstruct track routes or commuting times.

To test whether missing housing information matters, seventeen resale features describe trailing two-year area, floor and price distributions from 2017–2024 records [5], with a two-month lag and frozen validation origin. Sixteen building-stock features use a current property snapshot [6], without claiming historical availability. A separate CPU screen compares LightGBM [7] and an embedding MLP. Log-target CatBoost fits $\log y$ and restores currency predictions using Duan's smearing method [8]: the mean exponentiated training residual multiplies the exponentiated query prediction. This corrects retransformation using training data, not validation-label calibration.

IV. Experimental Design

A. Local validation and metrics

Chronological validation reflects the predominantly later competition period without assuming a pure time series. Each origin fits on earlier months and retains the complete scoring population. Models and rental-label histories remain fixed throughout its validation interval: no validation rents update later predictions. The recent quarter uses 141,076 training and 8,924 validation rows. Table II records cutoffs for all four windows. These overlapping windows have been repeatedly inspected during development and are not untouched final tests.

RMSE emphasises large deviations, MAE measures typical absolute error, and signed bias distinguishes systematic overprediction from underprediction. Error is prediction minus observation. For subgroup g, squared-error share is $\sum_{i \in g}(\widehat{y}_i - y_i)^2 / \sum_i (\widehat{y}_i - y_i)^2$. Counts accompany subgroup scores: high RMSE in a rare group can contribute less total error than moderate RMSE in a common group.

Historical exploration used scoring-window early stopping for CatBoost, with patience 100. Those results are descriptive references, not unbiased final comparisons. Subsequent temporal comparisons fix 1,600 trees per component and supply no outer scoring labels for early stopping. We keep the protocols separate: a changed tree-selection procedure must not be presented as a gain caused solely by a new feature. Neither procedure uses hidden competition labels.

B. Budgets, promotion and uncertainty

Selected CatBoost settings are 1,600 trees, learning rate 0.04, depth 8 and leaf L_2 regularisation 6, shared by deployment and fixed-budget validation. Alternative depths, sample weighting and targets are bounded checks, not exhaustive tuning; historical settings remain in the evidence ledger.

History shrinkage, the twelve-month half-life and trend damping are pragmatic choices held fixed subsequently, without independent optimisation. Four- and eight-component ensembles have equal per-component budgets but different total compute, so diversity and model count are not isolated.

Later exploratory experiments usually start with seed 42 on the recent quarter and sixteen-month window. Candidates replace the matching direct/residual pair in the frozen eight-component selected ensemble, or enter a predefined 25% blend. Budgets differ from full eight-component retraining. The later candidate acceptance criterion requires approximately 2 RMSE recent improvement plus a positive long-window gain before another seed and further periods. This operational threshold discourages selecting changes comparable to fitting noise; it is not a statistical significance level.

The CPU screen instead uses four one-seed components (two views, direct/residual), scoring 441.8429 recently, not the

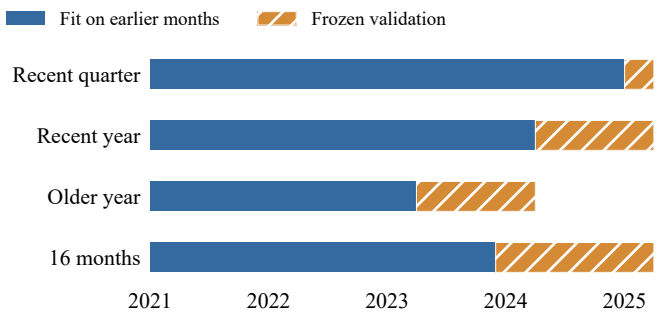


Fig. 3. Four local development windows. Each row fits on earlier months and scores the orange interval without updating rental-label histories. Windows overlap; they are not independent untouched tests. Exact cutoffs and sample sizes are in Table II.

TABLE I
EXPLORATORY CORE-INPUT REFERENCES, JANUARY–MARCH 2025 (SGD).

Historical core model	RMSE	MAE
Global median	844.60	691.33
Town/type median	546.81	425.11
Linear regression	523.20	373.09
Ridge	520.01	366.58
Decision tree	485.90	358.59
Random forest	470.19	348.52
Histogram boosting	463.36	336.75
CatBoost (core)	455.71	337.09

CatBoost uses scoring-window early stopping; representations and fitting budgets differ across learners. This is exploratory evidence.

frozen GPU reference. Log-target integration requires no loss across three windows and at least one two-point gain. Both screens fix 1,600 trees without outer-label early stopping; their distinct rules prohibit a retrospective common ranking.

Paired comparisons resample address clusters 1,000 times, reporting the 2.5th and 97.5th gain percentiles. Original keys use raw town/block/street strings, potentially splitting buildings. Canonicalised-key sensitivity retains all four gain directions: recent-quarter interval $[3.62, 5.85]$ and older-year $[-11.12, -8.59]$. These exploratory intervals preserve within-key association, not cross-month independence or adjustment for candidate search. GPU refits can vary; frozen predictions and hashes support auditability, not bit-identical training.

V. RESULTS AND INTERPRETATION

Fig. 3 shows the local evaluation spans. The adaptive reference averages direct and residual models over two seeds (four components); history diversity adds augmented-history counterparts (eight). The selected local-trend model changes only the normal-history half by adding four trend features. Stacking changes their combination weights, not the underlying fitted components.

A. Baseline and spatial information

In Table I, town/type medians outperform the global median, followed by gains from nonlinear trees under the tested

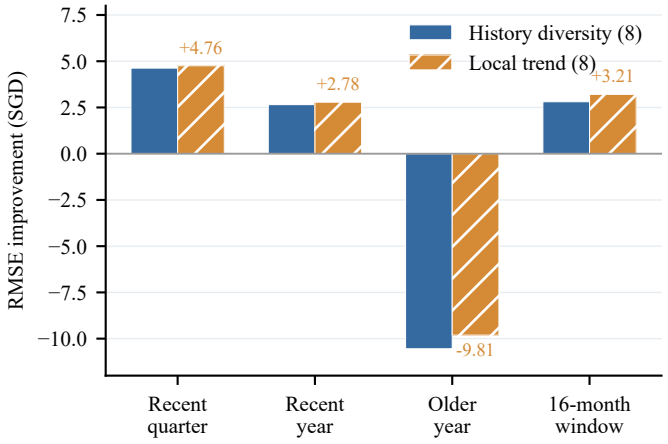


Fig. 4. RMSE improvement relative to the four-component adaptive reference. Positive bars mean lower error; both eight-component designs regress on the older year. Categories are separate, overlapping development windows.

representations and budgets. Historical spatial ablations give CatBoost core RMSE 455.71, block 454.19, block plus MRT 452.89, block plus malls 453.97, block plus schools 453.71, and all spatial features 452.62. Correlated address information and imperfect snapshots limit amenity interpretation.

The detailed-comparable two-seed model reaches 448.06 under historical early stopping. Later fixed-budget adaptive direct/residual averaging gives 445.52 on the same recent rows. These milestones use different protocols. Table II supplies the controlled fixed-budget evidence for subsequent selection.

B. Recent gains and an older-period regression

History diversity reduces adaptive-reference RMSE by 4.63 on the recent quarter, 2.65 on the recent year and 2.81 on sixteen months, but worsens the older year by 10.55. Adding local trends further reduces RMSE by 0.13 on the recent quarter, 0.13 on the recent year, 0.40 on sixteen months and 0.74 on the older year. The selected recipe therefore still loses 9.81 versus the adaptive reference on the older year. Its older bias is +82.72 SGD, compared with +8.89 for the adaptive reference. Improvement is conditional on market period.

The selected recent-quarter RMSE is 440.76, MAE 316.55 and bias +0.79 SGD. Its RMSE gain over the adaptive reference is 4.76, with a 95% paired address-cluster bootstrap interval of $[3.64, 5.87]$. On the older year the gain is -9.81, with interval $[-11.08, -8.59]$. Both use 1,000 resamples of the recorded predictions; these intervals describe the inspected windows rather than an untouched future period. Near-zero aggregate bias also leaves subgroup calibration unresolved.

We retain the selected model because the later-dated task favours recent conditions and it improves both recent and longer-horizon development windows. We accept eight components instead of four for those gains, while retaining the adaptive reference for robustness. Fig. 4 shows the unresolved older-period cost. There, MAE falls slightly from 455.20 to 454.38 while RMSE rises, directing attention to large

TABLE II
FIXED-BUDGET DEVELOPMENT RMSE (SGD); OVERLAPPING WINDOWS.

Validation interval	Train through	n	Adaptive (4)	Diversity (8)	Local trend (8)
Jan-Mar 2025	2024-12	8,924	445.52	440.89	440.76
Apr 2024-Mar 2025	2024-03	35,158	469.54	466.89	466.76
Apr 2023-Mar 2024	2023-03	35,882	580.54	591.08	590.35
Dec 2023-Mar 2025	2023-11	47,171	486.64	483.83	483.43

Training starts in January 2021 and ends at the listed cutoff. Four or eight components, each with 1,600 trees and seeds 42/2026; total compute differs. No scoring rents enter fitting, early stopping or within-window history updates.

residuals. Stale-history augmentation changes the training distribution and may overcompensate under a different regime; the observed errors do not establish that mechanism.

Frozen-component decomposition localises the trade-off. Older-year normal-history RMSE/bias are 579.85/+7.08 versus stale-history 624.08/+158.35; all canonical types regress against the adaptive reference. Recently, stale history instead scores 441.78 versus normal history 445.20. This compares prediction behaviour, not a causal augmentation effect, because view features differ. Subtracting half an earlier OOF origin's mean error gives older/recent RMSE 596.27/440.77: inherited constant bias correction does not resolve regime change.

C. Stacking and supplementary competition evidence

Four-head recent-window stacking improves local recent RMSE from 440.7629 to 440.6565, but its paired gain interval $[-0.0342, 0.2262]$ crosses zero. The eight-output version instead scores 440.8165. Earlier transfer tests are inconsistent; a sixteen-month stress check loses 0.538 with four inputs and 1.189 with eight. That check changes regime, fitting interval and horizon together, so it does not isolate recency weighting. Stacking remains an alternative rather than the default.

Two further weighting studies reuse the frozen component predictions. One varies the normal/stale-history weight linearly with forecast horizon; the other learns separate weights for address/type histories containing 0–4, 5–19 or at least 20 records. Both learn only from earlier validation windows and shrink weights toward 50:50. Each improves three of seven overlapping forward comparisons but worsens recent-quarter RMSE, to 441.08 and 441.09, respectively. Both also worsen the recent-year and sixteen-month results. These tests support retaining equal weights within the evaluated settings; neither horizon nor historical support provides a stable weighting rule across periods.

The competition provides supplementary public RMSE: 432.14881 for the selected model and 432.04286 for stacking. Two later four-component CPU probes score 432.70689 with building-stock features and 434.72053 with a market-normalized target; neither is promoted. Their seed count, device and composition differ from the selected ensemble, so public differences do not isolate feature effects. Public-subset membership and official rent labels are hidden; these scores cannot diagnose mechanisms or replace local evaluation. The submission ledger retains all probes, including those made despite failing local advancement rules.

TABLE III
RECENT ERRORS BY ACTUAL RENT (SGD); DIAGNOSTIC GROUPS ONLY.

Actual rent	n	RMSE	Bias	SSE (%)
$\leq$2500	1,458	788.1	+589.6	52.2
2501-3000	2,525	294.4	+47.3	12.6
3001-3500	2,823	276.2	-113.2	12.4
3501-4000	1,541	366.7	-252.8	12.0
>4000	577	568.9	-456.0	10.8

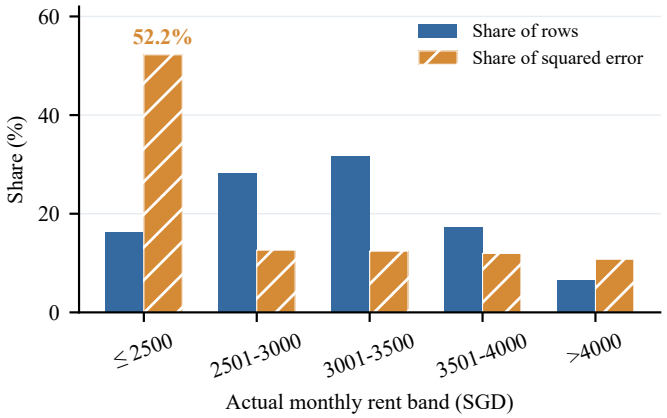


Fig. 5. Share of recent observations versus share of selected-model squared error. The lowest actual-rent band contributes disproportionately. Membership uses observed targets and is unavailable at prediction time.

An earlier keep-versus-remove sensitivity probe changes public RMSE by only 0.17 in a different model family. It supplies no convincing reason to declare flagged labels wrong. The selected model retains all labels; every local comparison retains its complete scoring population.

D. Where errors concentrate

Table III and Fig. 5 show that 1,458 rents at or below 2,500 SGD are 16.34% of recent rows but contribute 52.23% of squared error; the corresponding long-window share is 50.86%. Bias is prediction minus observation. This broad band includes plausible low-priced flats and is not an anomaly label. Low observed rents are overpredicted and high rents generally underpredicted. Conditioning on the realised target itself can contribute to this pattern; the plot alone cannot establish omitted contract conditions or incorrect labels as its cause.

TABLE IV
RECENT ERRORS BY OBSERVABLE FLAT TYPE (SGD).

Observed flat type	n	RMSE	Bias	SSE (%)
1-room	4	202.5	-27.8	0.0
2-room	242	408.9	+18.2	2.3
3-room	2,915	420.7	+11.9	29.8
4-room	3,201	436.1	0.0	35.1
5-room	2,100	462.8	-12.3	25.9
executive	462	506.7	-13.0	6.8

All 8,924 validation rows; bias is prediction minus observation. Groups are known at prediction time. The four one-room observations are descriptive only; small contributions may round to zero.

Table IV complements outcome-defined bands. Executive flats have recent RMSE 506.73 across 462 rows, yet only 6.84% of squared error. Three- and four-room groups together contribute 64.87%, partly reflecting prevalence. A narrower diagnostic event, actual rent below 65% of the frozen prediction, occurs in 2.992% of recent rows and contributes 32.47% of squared error. A feature-based detector has AUC 0.5761 and average precision 0.04161, versus prevalence 0.02992. This weak ranking signal supports neither reliable hard routing nor a claim that these events are completely unpredictable. Both prevalence and error magnitude matter when selecting a specialist population.

For this detector, a depth-4 CatBoost binary classifier uses 600 trees, learning rate 0.04 and $L_2 = 20$. It trains on 35,239 January–December 2024 transactions with earlier-origin OOF predictions, and is evaluated on January–March 2025. Inputs combine observable flat, location and amenity attributes, calendar month and the frozen base prediction; actual query rent only defines the evaluation label. This keeps the diagnostic distinct from the three-band expert classifier.

A simpler earlier-OOF town/type risk estimate gives recent AUC 0.5134 and average precision 0.03095, close to prevalence 0.02992; older-year AUC is 0.5176. Richer housing information is also tested against a distinct event: rent below 65% of a strictly-past adaptive address/type/area reference. This covers 282 recent rows (3.160%) and 32.598% of reference squared error. Adding housing features changes AUC from 0.65761 to 0.65835, with an incremental 95% address-cluster interval of $[-0.00625, 0.00740]$. A two-expert relative-rent mixture and its 25% blend score 453.993 and 442.223 versus CPU reference 441.843. Thus some ranking signal exists, but the added information does not yield useful routing. The different event definitions are not interchangeable or verified room-rental labels.

A separate reliability audit groups frozen predictions by historical support, recency and rent dispersion, using only records available at each validation origin. Dispersion is the address/type interquartile range divided by the median of market-normalised rent ratios, estimated with at least five records. In the recent quarter, the high-dispersion group (ratio above 0.25) contains 18.53% of observations but contributes 24.88% of squared error. Its RMSE exceeds that of the low-dispersion group in all four main windows, including within strata of support, flat type, month and area. Matched strata cover only part of the population, and the earliest supplementary annual window reverses the unstratified pattern. Support and recency associations are less stable after stratification. Historical dispersion therefore identifies elevated error risk, but does not establish label errors or show that stronger smoothing would improve prediction.

TABLE V
EXPLORATORY CANDIDATE COMPARISONS: RMSE (SGD).

Candidate	Use	Recent	16-month
Selected model	–	440.763	483.434
Nearest-comparable blend	B	441.842	482.198
Flat-type specialist blend	B	441.261	483.356
Rent-band mixture blend	B	441.962	482.488
Cross-fitted statistics	R	442.344	483.672
Strict-past distribution	R	440.641	483.299
COE indicators	R	440.877	483.434
Stock indicators	R	440.816	483.536
Matched-input XGBoost	R	441.881	485.244
OOF residual correction	C	440.922	483.340
Guarded Ridge blend	B	442.398	481.515
CPI	R	440.860	483.557
URA rental index	R	440.878	483.528
Corrected static MRT	R	440.631	483.418
MRT opening exposure	R	440.760	483.352
Monthly-available CPI+URA	R	440.791	483.478

B: 25% candidate blend; R: replacement of two selected components; C: 25% OOF residual correction. Candidates use one seed and differing budgets.

Large errors occur beyond rare executive flats. Contract terms, condition, renovation, household relationships or label problems remain possible explanations. Further work should establish a correctable, stable error pattern before fitting another specialist.

E. Exploratory extensions and negative evidence

Table V compares hypotheses under differing budgets. Nearest-comparable, specialist and soft-mixture blends can improve sixteen months while losing recently. Matched-input XGBoost worsens both windows; highly correlated errors limit the tested blend's complementarity. These findings favour the selected recipe within the tested budget.

Cross-fitting excludes a held-out inner fold from training statistics; outer validation uses only outer-training labels. Full-interval statistics worsen both windows, whereas strict-past dispersion replacement gains only about 0.12 and 0.14 RMSE. Changed training/inference support distributions limit generalisation of this finding. The isolated area/age composition-trend ablation provides essentially no recent gain and worsens the other main windows.

a) Parent–child reference accuracy: To determine which historical level might benefit from stronger smoothing, we compare parent and child references on identical high-dispersion validation rows. All references use origin-frozen histories and the same forecast-month market level, avoiding comparisons between unadjusted historical means and market-adjusted rents. Street/type references have higher RMSE than address/type references in all four core windows, providing

TABLE VI
DISPERSION-ADAPTIVE SHRINKAGE: PAIRED CPU RMSE (SGD).

Window	Target	Original	Candidate	Δ
Recent	Direct	446.090	445.555	-0.535
	Residual	447.295	447.522	$+0.227$
	Mean	445.759	445.656	-0.103
16 months	Direct	487.732	489.052	$+1.320$
	Residual	488.744	488.082	-0.662
	Mean	486.644	486.962	$+0.317$

Recent: Jan–Mar 2025 (8,924 rows); 16 months: Dec 2023–Mar 2025 (47,171 rows). Mean averages the two targets. Δ = candidate minus original; negative is better. One seed, matched validation rows and settings.

no support for generally favouring coarser groups. At the finest level, however, address/type references outperform address/type/area references by 2.79, 3.20, 4.34 and 3.85 SGD in the recent quarter, recent year, older year and sixteen-month window, respectively. The recent-quarter 95% address-cluster interval is $[-0.65, 6.76]$; the other three intervals exclude zero, although the windows overlap. In the audited data, these two finest levels use the same matched historical records. Their differences therefore arise from hierarchical smoothing rather than additional area-specific evidence. Low-dispersion controls do not show a consistent parent advantage, arguing against removing the finest level universally. These are gains in historical-reference accuracy, not CatBoost predictions: the model already receives both levels. The diagnostic thus motivates testing stronger shrinkage only at the finest level for high-dispersion cases, while leaving the remaining hierarchy unchanged.

b) Dispersion-adaptive shrinkage: Following this diagnostic, we conduct a controlled test of dispersion-adaptive historical smoothing. For address/type histories with at least five records and a market-normalised relative IQR above 0.25, we increase only the finest address/type/area shrinkage strength from 5 to 10 in Eq. (2). The modified premium and level are used as input features, while residual training targets and prediction offsets retain the original historical reference. All historical statistics use only permitted earlier training labels. We compare original and candidate recipes using CPU CatBoost with seed 42 and 1,600 trees, fitting direct and residual models on two validation windows (eight fits in total). This is a controlled component-level comparison rather than a full eight-model ensemble evaluation.

Table VI reports the results. Averaging direct and residual predictions yields a small RMSE improvement of 0.103 SGD in the recent quarter but a deterioration of 0.317 SGD over the 16-month window. Paired address-cluster bootstrap intervals for RMSE improvement are $[-0.535, 0.796]$ and $[-0.793, 0.126]$, respectively, both including zero. Thus, historical dispersion is associated with prediction difficulty, but this fixed increase in shrinkage does not demonstrate a consistent benefit.

Post-hoc analysis shows that the 16-month deterioration is not concentrated at longer forecast horizons: RMSE increases by 0.324 and 0.310 SGD in the first and last eight months, respectively. Direct prediction worsens in 14 of 16 months, while residual prediction improves in 11, partially offsetting the loss. Five-room flats account for 22.9% of observations but 60.0% of the net increase in squared error, although this does not establish a causal explanation. High-dispersion cases also show inconsistent monthly changes. Given the overlapping validation windows and single-seed CPU setting, we retain the original recipe rather than introduce subgroup-specific adjustments.

COE and stock features use preceding-month summaries and origin-frozen controls. Monthly lags approximate release timing, and stock membership is a supplied snapshot; neither materially improves prediction. Ridge exposes extrapolation instability when future numeric inputs exceed training ranges. Joint range clipping stabilises predictions, yet its blend loses recently while gaining over sixteen months. Multiple fields change together, preventing attribution to one variable.

Official CPI [9] activates two months after observation and URA [10] two months after quarter end. Origin-frozen indices are compared with a same-model control using releases available by each rental month. Revisions and rebasing are not reconstructed; URA private rents are only an HDB proxy. The monthly CPI+URA replacement remains slightly worse than the selected ensemble on both windows, despite improving its long-window frozen-input control.

The MRT audit finds 55 supplied rows marked open outside the verified operational calendar. Research variants verify TEL2–4, Punggol Coast and Hume dates against LTA releases (for example, TEL4 [11]), exclude future/unverified codes, deduplicate interchange names, and activate service from the first full month after opening. Origin-based validation admits dated openings announced before that origin; the monthly-access control admits operational service by the rental month. Supplied coordinates limit coverage, especially for LRT. Corrected static rail gains just 0.1322 and 0.0169 RMSE, with intervals crossing zero. Filtering and deduplication change jointly, so their contributions are not isolated. Dynamic access and opening exposure establish neither a reliable improvement nor a causal premium.

The index and rail studies comprise 36 formal GPU fits, excluding smoke checks. An identical-input seed-42 macro refit changes a replacement score by about 0.13; gains of this scale require caution. The selected recipe retains the supplied static MRT snapshot for comparability, not as a claim of operational access at every historical date.

Six additional interchange-access fits worsen recent RMSE by 0.084 under component replacement and 0.655 under a 25% blend. The latter improves the older year by 4.938, but its 585.41 RMSE still exceeds the adaptive reference's 580.54. Sixteen-month improvement is only 0.100. Line diversity therefore offers a period-dependent trade-off, not a stable replacement or causal transport effect.

Table VII tests a different target rather than another weight search. Log components improve the older-year mean by 2.33 RMSE but worsen the recent quarter by 0.28, failing the three-window rule. A separate 91-feature screen likewise reduces

TABLE VII
LOG-TARGET INTEGRATION UNDER THE ORIGINAL FEATURE WORKFLOW: RMSE (SGD).

Window	CPU ref. 4	Log 2	Mean 6
Jan–Mar 2025	441.84	445.17	442.12
Apr 2024–Mar 2025	467.30	469.65	467.27
Apr 2023–Mar 2024	585.84	585.50	583.52

One seed; reference: two views × direct/residual; log: two views; mean: six equally weighted components. Window populations match Table II; this is not the eight-component GPU comparison.

sixteen-month bias from +85.21 to +27.98 but recent RMSE worsens. Target transformation can change regime bias without providing uniform benefit.

In the four-component CPU screen, matched-input LightGBM scores 452.28 and an embedding MLP 471.61, versus reference 441.84. Replacing its normal-history direct/residual pair with resale or stock covariates gives 441.827 and 441.783; neither reaches the two-point screening threshold. These are bounded negative results, not proof that external information or other learners cannot help. Further duplicate-mean, market-normalized, deeper/asymmetric-tree and historical-only augmentation probes also fail advancement. The augmentation adds 4,357 attribute-matched public rental records [12] strictly before the scoring origin; no scoring-period targets enter fitting. Latest-vintage data and repeated development-window use limit these conclusions. Full compositions and numerical results remain in the repository.

We stop these candidate batches because gains are small, unstable across windows or comparable to refit variation. Exploratory public probes are recorded separately from model promotion. Further tuning would reuse the same development labels without resolving the older-period failure.

VI. LIMITATIONS AND REPRODUCIBILITY

A. Scope of the conclusions

Evaluation reuse. Temporal windows have been inspected repeatedly, several overlap, and public submissions add another selection channel. Selection on a noisy performance estimate can itself overfit [13]; bootstrap intervals do not repair this reuse. We have no private-label evaluation. The older-period regression prevents a claim of general dominance over the adaptive reference.

Incomplete observation. Unit floor, renovation and contract terms remain unobserved. Repeated building/type groups have constant supplied area, model and lease year, although resale records allow multiple unit attributes within those groups. The exact course join rule has not been recovered. Ambiguous flat identity and unusual rents establish neither label error nor a population irreducible-error floor. Predictions estimate recorded transactions, not a normative fair rent.

Information availability. Static MRT proxies, revised macro vintages, snapshot stock membership and approval/negotiation timing limit live deployment. Spatial and opening associations are not causal estimates.

Bounded search. Parameters and representations are not exhaustively explored. One-seed, two-component tests cannot rule out better full retraining. No uncertainty interval for an individual tenancy is supplied. Future work should identify stable observable subgroup bias in additional, preferably non-overlapping periods, then confirm promising corrections on later data and repeated seeds. Actual walking routes and commuting times require data beyond the tested station-distance and line-access proxies; these remain untested.

B. Implementation and reproducibility

The repository provides reusable code, the selected configuration, cumulative research decisions and score/provenance ledgers. Immutable course data, fitted models and detailed predictions remain local. Authorised readers can regenerate caches using the README; Appendix A maps the evidence.

An independent checkout/new Python 3.12 environment repeats eight 1,600-tree fits in 679.83 seconds on the RTX 4060, validating 50,000 predictions and model reloads. Recent refit RMSE is 440.6423 versus frozen 440.7629: fitting variation, not a method gain. CatBoost is pinned to 1.2.10. The maintained environment passes 74 tests, with four optional LightGBM tests skipped. Appendix A records reproduction scope; paper compilation needs no private caches.

Public-source target matching is quarantined from modeling. Its target-visible diagnostics are not held-out prediction results and are not used to train, tune or submit models; they establish neither a noise floor nor other teams' methods.

VII. CONCLUSION

For RQ1, canonical categories and validated joins establish coherent comparable groups; spatial ablations show modest incremental gains under the tested representation. For RQ2, market-adjusted comparables and history augmentation produce a selected recent RMSE of 440.76 and MAE of 316.55, with additional compute relative to the four-component reference. For RQ3, improvements transfer to two longer development windows but leave an older annual RMSE deficit of 9.81 unresolved. Low observed rents contribute disproportionately to squared error, while current feature-based detection is weak. The resulting lesson is a qualified modeling choice: retain useful recent-market information, document adverse transfer, and seek new observable evidence before further fine tuning.

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APPENDIX A
REPRODUCTION AND EVIDENCE MAP

Obtain the original course files under `data/raw/`, then run `python src/check_setup.py` and `python -m pytest -q`. The README supplies commands to regenerate temporal component predictions before feedback studies. Cached predictions and models are not distributed. Compiling the paper needs committed sources and aggregate assets; rebuilding evidence needs course data and caches. Full-budget selected-model training is verified in an independent checkout and new virtual environment on the same physical machine. GPU refits differ numerically; frozen hashes and metric checks prevent silently replacing old evidence. A new run requires a reviewed manifest and corresponding report update. Cross-device verification is a separate team task.

Fixed-budget comparisons use `src/run_experiments.py` with the `--compare-candidates` option. The selected ensemble is generated by `src/train_final_model.py` with the recorded configuration; research uses `src/feedback_experiments.py`, including `--families public` for indices and rail. These are abbreviated entry points; use the complete README commands for exact windows and options.

Supplementary CPU modules are under `src/research/`, with a separate `requirements-research.txt`. Their aggregate evidence is in `reports/november_cpu_comparisons_20261102.csv` and `reports/november_log_integration_20261102.csv`.
The consolidated research record preserves protocols and unsuccessful probes. The shrinkage study uses `src/research/dispersion_shrinkage.py`;
saved scores and paired uncertainty checks are under `artifacts/dispersion_shrinkage_cpu_v1/`.
The read-only module `src/research/analyze_shrinkage_errors.py` produces the monthly and subgroup evidence under `artifacts/dispersion_shrinkage_attribution_v1/`. These diagnostics reuse predictions without additional model fitting. For clarity, the historical core-baseline settings are listed in Table VIII. They are bounded references, not the outcome of an exhaustive or common-compute search. The selected ensemble's settings and history coefficients are given in the main text.

TABLE VIII
PRINCIPAL HISTORICAL BASELINE SETTINGS; PREPROCESSING IS DESCRIBED IN THE MAIN TEXT.

Model	Recorded settings
Linear/Ridge	Ordinary least squares; Ridge $\alpha = 10$
Decision Tree	Maximum depth 18; minimum leaf size 20
Random Forest	180 trees; maximum feature fraction 0.7; minimum leaf size 3
Histogram boosting	250 iterations; learning rate 0.08; 31 leaves; $L_2 = 2$

TABLE IX
EVIDENCE LOCATIONS; DETAILED LOCAL PREDICTIONS CAN BE REGENERATED.

Evidence	Location
Baseline/spatial results	`reports/experiment_results.csv`
Fixed-budget windows	`reports/temporal_summary.csv`
Candidate comparisons	`reports/feedback_results.csv`
Stacking transfer	`reports/stacking_results.csv`
Actual public scores	`reports/kaggle_submissions.csv`
Selected recipe	`configs/model.json`
Model provenance	`reports/final_model_metadata.json`
Research decisions	`docs/PROGRESS.md`

APPENDIX B
TEAM WORKLOAD STATEMENT – TO BE COMPLETED

Reserved for completion before final submission. Responsibilities and contribution proportions have not been assigned in this draft.

TABLE X
WORKLOAD PLACEHOLDER; NO CONTRIBUTIONS ARE ASSERTED.

Member	Student ID	Responsibilities
Han Linxuan	A0350911M	To be completed
Sun Shenghe	A0354849L	To be completed
Zhang Yuhong	A0356795H	To be completed
Zhang Zhexuan	A0355290A	To be completed

APPENDIX C
AI ASSISTANCE

AI tools, including Claude and Codex, assisted with discussion of experimental ideas, code development and checks, interpretation of recorded results, and drafting and editing this report. Numerical results derive from executed experiments and saved evidence; proposed analyses are identified separately. This statement documents assistance; it does not assert that team review or workload attribution has been completed.